bilalhusain126 Add Deep Picard Iteration solver for XVA BSDE

81e60e8 · 3 hours ago History

XVA (CVA + FVA) via Deep Picard Iteration

This notebook solves the nonlinear XVA BSDE for a caplet portfolio using the dim-fbsde solver.

The BSDE is:

$$-dXVA_t = f(t, F(t), XVA_t)\,dt - Z_t \cdot dW_t^Q, \quad XVA_T = 0$$

where the driver f aggregates CVA cost, FVA funding recursion, and intensity discounting. The nonlinearity arises from the funding terms $(\bar{V} - XVA - C)^{\pm}$.

In [1]:

```
import sys
import os

# Add paths for local packages
sys.path.insert(0, "..")

import numpy as np
import matplotlib.pyplot as plt

from plot_style import setup_publication_style
setup_publication_style()
```

```
setup_publication_style()
%config InlineBackend.figure_format = 'retina'

from Caplet_portfolio_data import example_build_problem
from xva_equation import XVACapletEquation, solve_xva
```

1. Build Market Data & Simulate Forward Paths

We start with a small problem (d=10 forwards, T=5y, monthly steps) for quick iteration.

In [2]:


```
    seed=42,
    threshold_H=5000,
)

data = problem.data
print(f"Forwards: d={meta['d']}")
print(f"Portfolio: M={meta['M']} caplets")
print(f"Paths: {data.X.shape[0]}, Time steps: {data.X.shape[1]}'
print(f"Horizon: T={meta['T_max']}y, dt={meta['dt']:.4f}")
print(f"Counterparty: lambda={meta['lambda_C']}, R={meta['recove
print(f"Funding: s_b={meta['funding_spread_b']}, s_l={meta['func
```

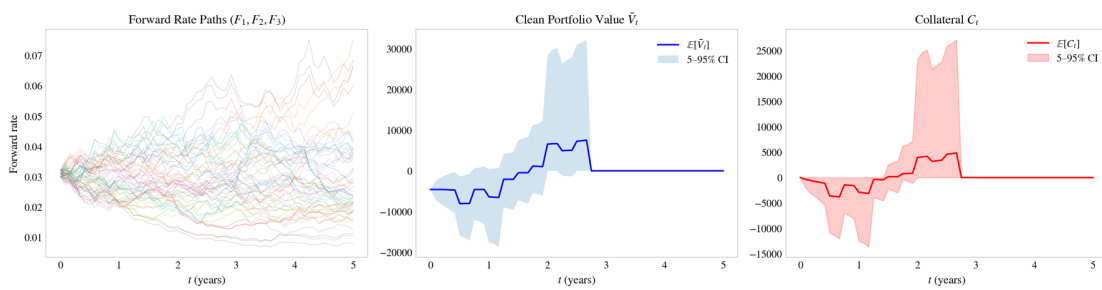
```
Forwards: d=10
Portfolio: M=50 caplets
Paths: 5000, Time steps: 61
Horizon: T=5.0y, dt=0.0833
Counterparty: lambda=0.015, R=0.4
Funding: s_b=0.01, s_l=0.0
```

2. Inspect Simulated Data

In [3]:

```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# Forward rate paths (first 3 forwards, 20 paths)
t = data.t
for i in range(min(3, meta['d'])):
    ax = axes[0]
```

```
    for p in range(20):
        ax.plot(t, data.X[p, :, i], alpha=0.3, lw=0.5)
    ax.set_title(r"Forward Rate Paths ($F_1, F_2, F_3$)")
    ax.set_xlabel(r"$t$ (years)")
    ax.set_ylabel(r"Forward rate")

# Clean portfolio value
axes[1].plot(t, np.mean(data.Vbar, axis=0), 'b-', lw=1.5, label=
axes[1].fill_between(t,
    np.percentile(data.Vbar, 5, axis=0),
    np.percentile(data.Vbar, 95, axis=0),
    alpha=0.2, label='5–95% CI')
axes[1].set_title(r"Clean Portfolio Value $\bar{V}_t$")
axes[1].set_xlabel(r"$t$ (years)")
axes[1].legend()

# Collateral
axes[2].plot(t, np.mean(data.C, axis=0), 'r-', lw=1.5, label=r'$
axes[2].fill_between(t,
    np.percentile(data.C, 5, axis=0),
    np.percentile(data.C, 95, axis=0),
    alpha=0.2, color='red', label='5–95% CI')
axes[2].set_title(r"Collateral $C_t$")
axes[2].set_xlabel(r"$t$ (years)")
axes[2].legend()

plt.tight_layout()
plt.show()
```



3. Solve XVA BSDE

In [4]:

```
out = solve_xva(
    d=10,
    T_max=5.0,
    dt=1.0 / 12.0,
    n_paths=3000,
    picard_iterations=10,
    epochs=5,
    learning_rate=1e-4,
```

```
    hidden_dims=[256, 256],
    threshold_H=1e6,
    seed=42,
    device="cpu",
)

print(f"\nXVA_0 = {out['XVA_0']:.4f}")
```

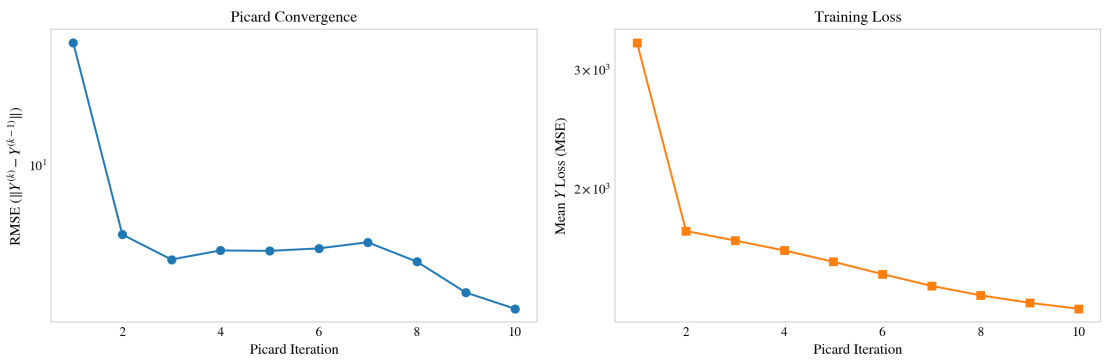
```
2026-03-04 15:36:54 - dim_fbsde.solvers.uncoupled - Starting Unco
upled Solver (gradient)...
2026-03-04 15:36:56 - dim_fbsde.solvers.uncoupled - Picard Iter
1/10: Error=5.4010e+01 | Mean Y_Loss=3.2886e+03
2026-03-04 15:36:58 - dim_fbsde.solvers.uncoupled - Picard Iter
2/10: Error=3.8459e+00 | Mean Y_Loss=1.7302e+03
2026-03-04 15:37:00 - dim_fbsde.solvers.uncoupled - Picard Iter
3/10: Error=2.7283e+00 | Mean Y_Loss=1.6746e+03
2026-03-04 15:37:02 - dim_fbsde.solvers.uncoupled - Picard Iter
4/10: Error=3.0905e+00 | Mean Y_Loss=1.6192e+03
2026-03-04 15:37:04 - dim_fbsde.solvers.uncoupled - Picard Iter
5/10: Error=3.0718e+00 | Mean Y_Loss=1.5580e+03
2026-03-04 15:37:07 - dim_fbsde.solvers.uncoupled - Picard Iter
6/10: Error=3.1789e+00 | Mean Y_Loss=1.4938e+03
2026-03-04 15:37:09 - dim_fbsde.solvers.uncoupled - Picard Iter
7/10: Error=3.4581e+00 | Mean Y_Loss=1.4350e+03
2026-03-04 15:37:11 - dim_fbsde.solvers.uncoupled - Picard Iter
8/10: Error=2.6482e+00 | Mean Y_Loss=1.3892e+03
2026-03-04 15:37:13 - dim_fbsde.solvers.uncoupled - Picard Iter
9/10: Error=1.7306e+00 | Mean Y_Loss=1.3545e+03
2026-03-04 15:37:15 - dim_fbsde.solvers.uncoupled - Picard Iter 1
0/10: Error=1.3842e+00 | Mean Y_Loss=1.3272e+03

XVA_0 = -151.5506
```

4. Convergence Analysis

In [5]:

```
history = out['result']['history']

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Picard iteration error
errors = [h['error'] for h in history]
axes[0].semilogy(range(1, len(errors) + 1), errors, 'o-')
axes[0].set_xlabel('Picard Iteration')
axes[0].set_ylabel(r'RMSE ($\|Y^{(k)} - Y^{(k-1)}\|$)')
axes[0].set_title('Picard Convergence')

# Y training loss
y_losses = [np.mean(h['y_loss']) for h in history]
axes[1].semilogy(range(1, len(y_losses) + 1), y_losses, 's-', co
```

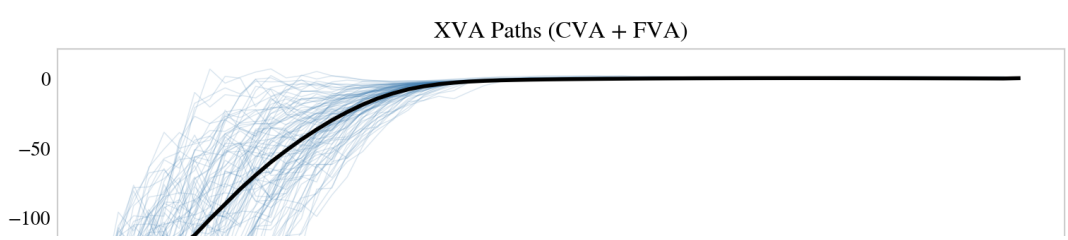
```
axes[1].set_xlabel('Picard Iteration')
axes[1].set_ylabel(r'Mean $Y$ Loss (MSE)')
axes[1].set_title('Training Loss')

plt.tight_layout()
plt.show()
```

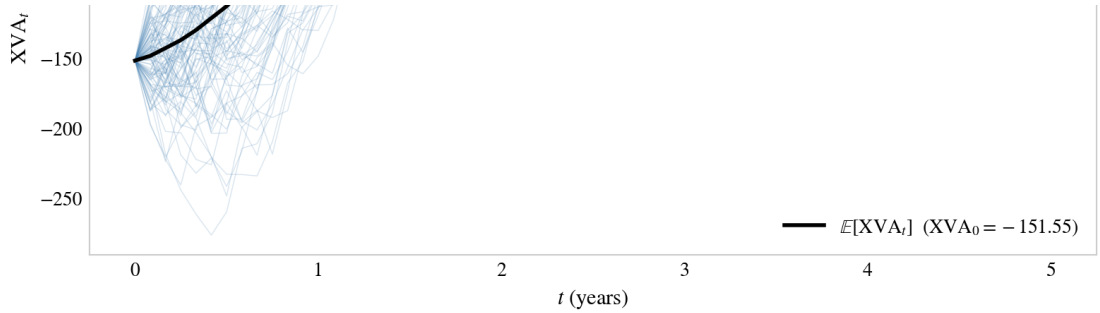


5. XVA Paths

In [7]:

```
Y = out['result']['Y']  # shape [M, N+1, 1]
time = out['result']['time']

fig, ax = plt.subplots(figsize=(8, 4))

# Plot sample paths
n_plot = 100
for p in range(n_plot):
    ax.plot(time, Y[p, :, 0], alpha=0.2, lw=0.5, color='steelblu

# Mean XVA path
ax.plot(time, Y[:, :, 0].mean(axis=0), 'k-', lw=2,
        label=r'$\mathbb{E}[\mathrm{XVA}_t]$' + f'  ($\\mathrm{{

ax.set_xlabel(r'$t$ (years)')
ax.set_ylabel(r'$\mathrm{XVA}_t$')
ax.set_title(r'XVA Paths (CVA + FVA)')
ax.legend()
plt.tight_layout()
plt.show()
```





In [ ]:

```

```